

CSCI 240 -- Review for Exam 1

Textbook

- Chapters 1 to 7

Review

- HW 1
- PAs 1 to 4
- In-class exercise 1 and pop quiz 1
- Design Goals -- robustness, adaptability, and reusability
- OOD Principles -- abstraction, encapsulation, and modularity
- Functions/Methods -- set up and use functions/methods
- Classes -- components of a class, class access, set up and use classes
- Inheritance and Polymorphism -- base and derived classes, static vs. dynamic binding
- Analysis of Algorithms -- the seven functions, analyze running times, different common growth rates, big-Oh, big-Omega, and big-Theta, summations
- Proof or Justification Techniques -- by example, contra-positive, contradiction, and induction
- Recursion -- advantages and disadvantages, linear recursion, binary recursion, and tail recursion
- Arrays -- static vs. dynamic arrays, array operations
- Linked Lists -- node, head and tail, header and trailer nodes, singly linked lists, doubly linked lists, traverse, insert, and delete
- Linked Lists -- arrays vs. linked lists, singly linked lists vs. doubly linked lists, header and trailer nodes, operations, advantages and disadvantages, performance
- Stacks -- the Stack ADT, array-based implementation, performance, evaluate postfix expressions
- Queues -- the Queue ADT, circular array implementation, linked list implementation, performance
- Deques -- the Deque ADT, performance
- Vectors -- the Vector ADT, performance
- Lists -- the List ADT, performance
- Sequences -- the Sequence ADT, performance
- Positions and Iterators -- Position ADT, Iterator ADT, why needed

Some Review Questions/Exercises

- 1) Write a C++ function or a Java method that accepts a positive integer n and returns true if n is a power of 2.
 - 2) Which principle(s) can be used to achieve the design goal of adaptability?
 - 3) Analyze running time for the following pseudocode:

```
for i = 1 to 10
    for j = 1 to i
        add 1 to count
```
 - 4) Come up with the name of an algorithm for each of the seven functions; for example, binary search has a runtime of $O(\log n)$
 - 5) Given a pointer/reference current that refers to a certain node in a singly linked list, describe the process to remove that node.
 - 6) Given a pointer/reference current that refers to a certain node in a singly linked list, describe the process to add a new node after that node.
 - 7) Draw a simple UML class diagram showing relationships between some classes (including has-a and is-a relationships).
 - 8) Set up a class like GameEntry.
 - 9) Come up with a recursive solution to sum all values from 1 to n .
 - 10) Given an array, insert a value to the array at index i .
 - 11) Provide an algorithm and evaluate the following postfix expression: $8\ 3\ /\ 4\ +$
 - 12) Explain how we can use a deque to implement a queue.
 - 13) What are two possible setups for the pop() operation? Which one is better?
 - 14) Why is it more difficult to implement a queue with an array than implementing a stack with an array?
 - 15) Insert a node before p or after node p from a doubly linked list.
 - 16) Remove node p from a doubly linked list.
 - 17) What are some differences between a Vector and a List?
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